

$$3d^1$$

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$$1 \quad \left| 3d_{z^2}^\alpha 3d_{xz}^0 3d_{yz}^0 3d_{xy}^0 3d_{x^2-y^2}^0 \right\rangle$$

Here are the CSFs:

$$\begin{aligned} |1, 1/2, 1/2\rangle &= \left| 3d_{xz}^\alpha 3d_{yz}^0 3d_{xy}^0 3d_{z^2}^0 3d_{x^2-y^2}^0 \right\rangle \\ |2, 1/2, 1/2\rangle &= \left| 3d_{xz}^0 3d_{yz}^\alpha 3d_{xy}^0 3d_{z^2}^0 3d_{x^2-y^2}^0 \right\rangle \\ |3, 1/2, 1/2\rangle &= \left| 3d_{xz}^0 3d_{yz}^0 3d_{xy}^\alpha 3d_{z^2}^0 3d_{x^2-y^2}^0 \right\rangle \\ |4, 1/2, 1/2\rangle &= \left| 3d_{xz}^0 3d_{yz}^0 3d_{xy}^0 3d_{z^2}^\alpha 3d_{x^2-y^2}^0 \right\rangle \\ |5, 1/2, 1/2\rangle &= \left| 3d_{xz}^0 3d_{yz}^0 3d_{xy}^0 3d_{z^2}^0 3d_{x^2-y^2}^\alpha \right\rangle \\ |6, 1/2, -1/2\rangle &= \left| 3d_{xz}^\beta 3d_{yz}^0 3d_{xy}^0 3d_{z^2}^0 3d_{x^2-y^2}^0 \right\rangle \\ |7, 1/2, -1/2\rangle &= \left| 3d_{xz}^0 3d_{yz}^\beta 3d_{xy}^0 3d_{z^2}^0 3d_{x^2-y^2}^0 \right\rangle \\ |8, 1/2, -1/2\rangle &= \left| 3d_{xz}^0 3d_{yz}^0 3d_{xy}^\beta 3d_{z^2}^0 3d_{x^2-y^2}^0 \right\rangle \\ |9, 1/2, -1/2\rangle &= \left| 3d_{xz}^0 3d_{yz}^0 3d_{xy}^0 3d_{z^2}^\beta 3d_{x^2-y^2}^0 \right\rangle \\ |10, 1/2, -1/2\rangle &= \left| 3d_{xz}^0 3d_{yz}^0 3d_{xy}^0 3d_{z^2}^0 3d_{x^2-y^2}^\beta \right\rangle \end{aligned}$$

The Ground State CSFs are 4 9

$$\langle 2, \frac{1}{2}, \frac{1}{2} | L_x | 4, \frac{1}{2}, \frac{1}{2} \rangle = -\sqrt{3}I$$

$$\Delta g_{xx}$$

$$-6\zeta\Delta_{4-2}^{-1}$$

$$\langle 1,\frac{1}{2},\frac{1}{2}|L_y|4,\frac{1}{2},\frac{1}{2}\rangle=\sqrt{3}I$$

$$\Delta g_{yy}$$

$$-6\zeta\Delta_{4-1}^{-1}$$